Quiz 2

This is an individual assignment.

For the analytical problems below, you can:

1. write your answers in markdown cells with $\LaTeX$, *or*
2. Embed pictures with your handwritten answers inside the Notebook, *or*
3. ```push``` a single pdf to your repository with all handwritten answers.

Exercise 1 $3 points$

Suppose you have a training set with N data points $\{x_i\}_{i=1}^N$, where $x_i \in \mathbb{R}$. Assume the samples are independent and identically distributed $i.i.d.$, and each sample is drawn from a Gaussian random variable with *known* mean μ and *unknown* variance σ^2. The probability density function is defined as:

$$p(x|\mu,\sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}}\exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$$

where $\mu, \sigma > 0$.

Moreover, consider the Inverse Gamma as the prior probability on the hyperparameter σ^2,

$$p(\sigma^2|\alpha,\beta) = \frac{\beta^\alpha}{\Gamma(\alpha)}x^{-\alpha-1}\exp\left(-\frac{\beta}{x}\right)$$

where $\alpha, \beta > 0$.

Answer the following questions:

1. $1point$ **Derive the maximum likelihood estimate MLE for the parameter σ^2. Show your work.**

EX1.

MLE: $\mathcal{L}(\sigma^2) = \prod_{i=1}^{N} p(x_i | \mu, \sigma^2) = \prod_{i=1}^{N} \frac{1}{\sqrt{2\pi\sigma^2}} e^{\left(\frac{(x_i-\mu)^2}{-2\sigma^2}\right)}$

$$= \left(\frac{1}{\sqrt{2\pi\sigma^2}}\right)^N \cdot e^{\left(\frac{\sum^{N}(x_i-\mu)^2}{-2\sigma^2}\right)}$$

$$\log_e \mathcal{L}(\sigma^2) = N \log_e\left(\frac{1}{\sqrt{2\pi\sigma^2}}\right) + \frac{1}{-2\sigma^2}\left(\sum_{i=1}^{N}(x_i-\mu)^2\right)$$

$$= -N \cdot \frac{1}{2}\left(\log_e(2\pi) + \log_e(\sigma^2)\right) - \frac{1}{2\sigma^2}\left(\sum_{i=1}^{N}(x_i-\mu)^2\right)$$

$$\frac{\partial \log_e \mathcal{L}(\sigma^2)}{\partial \sigma^2} = 0 = -\frac{N}{2} \cdot \frac{1}{\sigma^2} - \frac{-1}{2(\sigma^2)^2}\left(\sum_{i=1}^{N}(x_i-\mu)^2\right)$$

$$\Rightarrow \frac{N}{2\sigma^2} = \frac{1}{2(\sigma^2)^2}\left(\sum_{i=1}^{N}(x_i-\mu)^2\right) \Rightarrow \sigma^2 = \frac{1}{N}\left(\sum_{i=1}^{N}(x_i-\mu)^2\right)$$ #

2. $1point$ **Derive the maximum a posteriori MAP estimate for the parameter σ^2. Show your work.**

MAP: $p(\sigma^2|X) \propto p(X|\sigma^2)\cdot p(\sigma^2) \Rightarrow \log_e p(\sigma^2|X) = \log_e p(X|\sigma^2) + \log p(\sigma^2) + \text{constant}$

$\log p(\sigma^2) = \alpha\log\beta - \log\Gamma(\alpha) - (\alpha+1)\log(\sigma^2) - \frac{\beta}{\sigma^2}$, $\log_e p(X|\sigma^2) = \frac{-N}{2}\log(\sigma^2) - \frac{1}{2\sigma^2}\sum(x_i-\mu)^2 + \text{const}$

$$\log_e p(\sigma^2|X) = \frac{-N}{2}\log(\sigma^2) - \frac{1}{2\sigma^2}\sum_{i=1}^{N}(x_i-\mu)^2 - (\alpha+1)\log(\sigma^2) - \frac{\beta}{\sigma^2} + \text{const}$$

$$= \left(\frac{-N-2\alpha-2}{2}\right)\log(\sigma^2) - \frac{1}{2\sigma^2}\left(\sum_{i=1}^{N}(x_i-\mu)^2 + 2\beta\right) + \text{const}$$

$$\frac{\partial \log_e p(\sigma^2|X)}{\partial \sigma^2} = \left(\frac{-N-2\alpha-2}{2\sigma^2}\right) + \frac{1}{2(\sigma^2)^2}\left(\sum_{i=1}^{N}(x_i-\mu)^2 + 2\beta\right) = 0$$

$$\Rightarrow \frac{N+2\alpha+2}{2\sigma^2} = \frac{1}{2\sigma^2\sigma^2}\left(\sum_{i=1}^{N}(x_i-\mu)^2 + 2\beta\right)$$

$$\Rightarrow \sigma^2_{MAP} = \frac{\sum_{i=1}^{N}(x_i-\mu)^2 + 2\beta}{N+2\alpha+2}$$

3. $1 point$ **Show that the Inverse Gamma forms a conjugate prior with the Gaussian likelihood, and write down the pseudo-code for the online update of the prior parameters.**

Solution:

To show that Inverse Gamma is a conjugate prior, we need to demonstrate that the posterior has the same functional form as the prior.

From part 2, the log-posterior was:

$$\log p(\sigma^2|X) = -\left(\frac{N}{2} + \alpha + 1\right)\log(\sigma^2) - \frac{1}{\sigma^2}\left(\frac{1}{2}\sum_{i=1}^{N}(x_i-\mu)^2 + \beta\right) + \text{const}$$

This can be rewritten as:

$$p(\sigma^2|X) \propto (\sigma^2)^{-(\alpha'+1)}\exp\left(-\frac{\beta'}{\sigma^2}\right)$$

where:

- $\alpha' = \alpha + \frac{N}{2}$

- $\beta' = \beta + \frac{1}{2}\sum_{i=1}^{N}(x_i - \mu)^2$

This is exactly the form of an Inverse Gamma distribution with updated parameters (α', β'). Therefore, Inverse Gamma is indeed a conjugate prior for the Gaussian likelihood with unknown variance.

Online Update Pseudocode:

```

Initialize: $\alpha_0$, $\beta_0$ # prior parameters

For each new data point $x_i$:

# Update parameters

$\alpha_1 = \alpha_0 + 1/2$

$\beta_1 = \beta_0 + x_i - \mu^2/2$

# Updated parameters become new prior

$\alpha_0 = \alpha_1$

$\beta_0 = \beta_1$

# Current MAP estimate

$\sigma^2$_MAP = $\beta_1/\alpha_1 + 1$

```

The key insight is that each new observation updates the prior parameters incrementally, allowing for sequential Bayesian inference without storing all previous data points.

Exercise 2 $3 points$

Motivation:

> Waste minimization is a critical aspect of sustainable productionpractices, particularly in the manufacturing of consumer goods. As theglobal demand for products continues to rise, so does the generation ofwaste, posing significant environmental and economic challenges. Theproduction of consumer goods involves numerous stages, from rawmaterial extraction to manufacturing, packaging, and distribution. Ateach step, there is the potential for waste generation, leading to resourcedepletion, pollution, and increased costs. Addressing this issue andimplementing effective waste minimization strategies is essential tomitigate the environmental impact, conserve resources, and optimizeproduction processes, all while maintaining profitability and meetingconsumer expectations.

Suppose your company is in the business of manufacturing consumer goods productsand that you are interested in minimizing waste production. Instead of producing alarge number of items at any given time, design a strategy to determine how manyproducts should be manufactured at any given time to minimize waste production.

1. $1 point$ **The dataset $X = \{x_i\}_{i=1}^{N}$ is comprised of N samples x_i which represent the "number of products to be produced in a given interval of time". Which random variable RV best describes this data $assumme\$x_i\$'s are i.i.d.$? For your choice of RV, write down the observed data likelihood, $\mathcal{L}$.**

__*Hint: select one from the most popular discrete RVs.*__

Solution:

Since x_i represents the "number of products to be produced in a given interval of time", we are dealing with count data over fixed time intervals. The most appropriate discrete random variable for this scenario is the **Poisson distribution**.

The Poisson distribution is ideal because:

- It models the number of events $product production$ occurring in a fixed interval

- It assumes events occur independently

- It's commonly used for manufacturing and production processes

For a Poisson distribution with parameter $\lambda > 0$:

$$p(x_i|\lambda) = \frac{\lambda^{x_i} e^{-\lambda}}{x_i!}$$

The observed data likelihood for N i.i.d. samples is:

$$\mathcal{L}(\lambda) = \prod_{i=1}^{N} p(x_i|\lambda) = \prod_{i=1}^{N} \frac{\lambda^{x_i} e^{-\lambda}}{x_i!}$$

$$= \frac{\lambda^{\sum_{i=1}^{N} x_i} e^{-N\lambda}}{\prod_{i=1}^{N} x_i!}$$

2. $1 point$ **Solve for the parameter of the observed data likelihood using the MLE approach.**

Solution:

From part 1, the likelihood function is:

$$\mathcal{L}(\lambda) = \frac{\lambda^{\sum_{i=1}^{N} x_i} e^{-N\lambda}}{\prod_{i=1}^{N} x_i!}$$

Taking the log-likelihood:

$$\ell(\lambda) = \log \mathcal{L}(\lambda) = \sum_{i=1}^{N} x_i \log \lambda - N\lambda - \log \left(\prod_{i=1}^{N} x_i! \right)$$

$$= \left(\sum_{i=1}^{N} x_i \right) \log \lambda - N\lambda + \text{const}$$

To find the MLE, we take the derivative with respect to λ and set it to zero:

$$\frac{\partial \ell}{\partial \lambda} = \frac{\sum_{i=1}^{N} x_i}{\lambda} - N = 0$$

Solving for λ:

$$\frac{\sum_{i=1}^{N} x_i}{\lambda} = N$$

$$\lambda = \frac{\sum_{i=1}^{N} x_i}{N}$$

Therefore, the MLE estimate is:

$$\hat{\lambda}_{MLE} = \frac{1}{N} \sum_{i=1}^{N} x_i = \bar{x}$$

The MLE for the Poisson parameter is simply the sample mean, which makes intuitive sense as λ represents the average rate of production.

3. $1 point$ **Introduce a prior probability for the parameter of the data likelihood defined in 1 such that it forms a conjugate prior relationship, and justify your choice. Derive the solution for the parameter using the MAP approach.**

Solution:

For the Poisson likelihood, the **Gamma distribution** forms a conjugate prior for the parameter λ.

Justification: The Gamma distribution is the natural conjugate prior for the Poisson likelihood because:

- Both involve exponential terms with λ

- The posterior will maintain the same functional form as the prior
- It's mathematically tractable and widely used in Bayesian analysis

The Gamma prior is defined as:

$$p(\lambda|\alpha,\beta) = \frac{\beta^\alpha}{\Gamma(\alpha)}\lambda^{\alpha-1}e^{-\beta\lambda}$$

where $\alpha > 0\ shape parameter$ and $\beta > 0\ rate parameter.$

MAP Derivation:

Using Bayes' theorem, the posterior is:

$$p(\lambda|X) \propto p(X|\lambda) \cdot p(\lambda)$$

From parts 1 and 2:

$$p(X|\lambda) \propto \lambda^{\sum_{i=1}^N x_i} e^{-N\lambda}$$

Combined with the Gamma prior:

$$p(\lambda|X) \propto \lambda^{\sum_{i=1}^N x_i} e^{-N\lambda} \cdot \lambda^{\alpha-1}e^{-\beta\lambda}$$

$$= \lambda^{\sum_{i=1}^N x_i + \alpha - 1} e^{-(\beta+N)\lambda}$$

This is a Gamma distribution with updated parameters:

- $\alpha' = \alpha + \sum_{i=1}^N x_i$
- $\beta' = \beta + N$

For a Gamma distribution, the mode $MAPestimate$ is:

$$\hat{\lambda}_{MAP} = \frac{\alpha' - 1}{\beta'} = \frac{\alpha + \sum_{i=1}^{N} x_i - 1}{\beta + N}$$

Exercise 3 $3points$

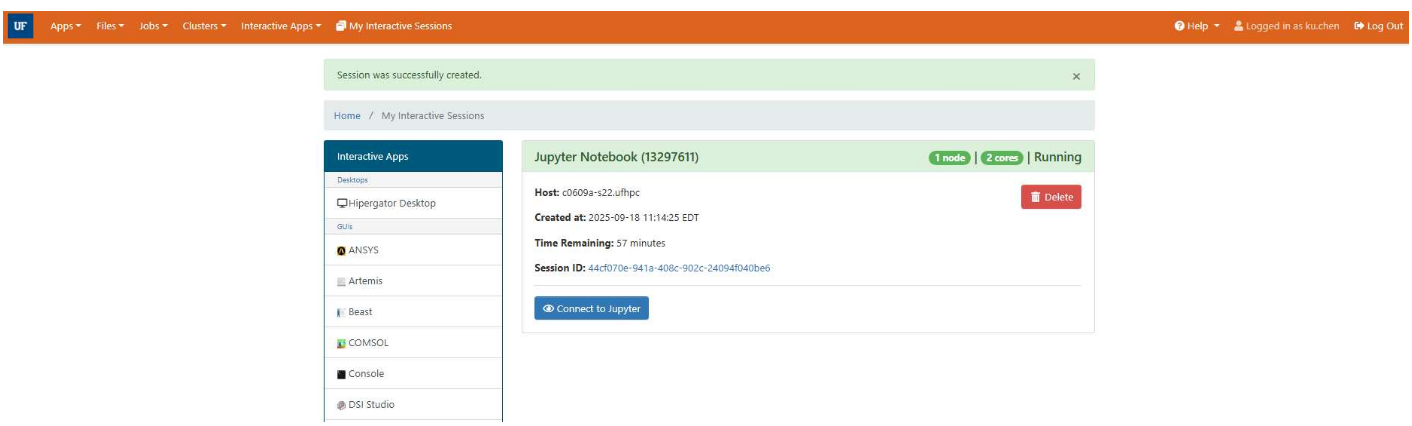
In this question, you will practice how to use HiPerGator and Git and GitHub to maintain your code.

1. $0.5points$ **Open "Open On-Demand"

$$ood.rc.ufl.edu$$

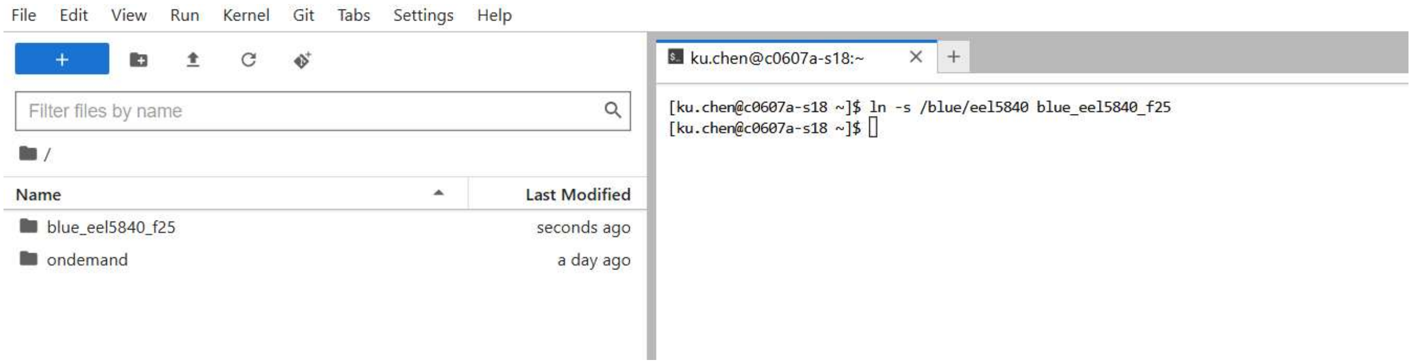
$ood.rc.ufl.edu$ and create an interactive jupyter session with the following specifications: 2 CPU, 2 GPU and 16 GB of RAM. Attach a screenshot of your jupyter notebook card, your gatorlink should be visible $examplebelow$.**

```
from IPython.display import Image
Image('figures/ood_job_request.png', width=900)
```



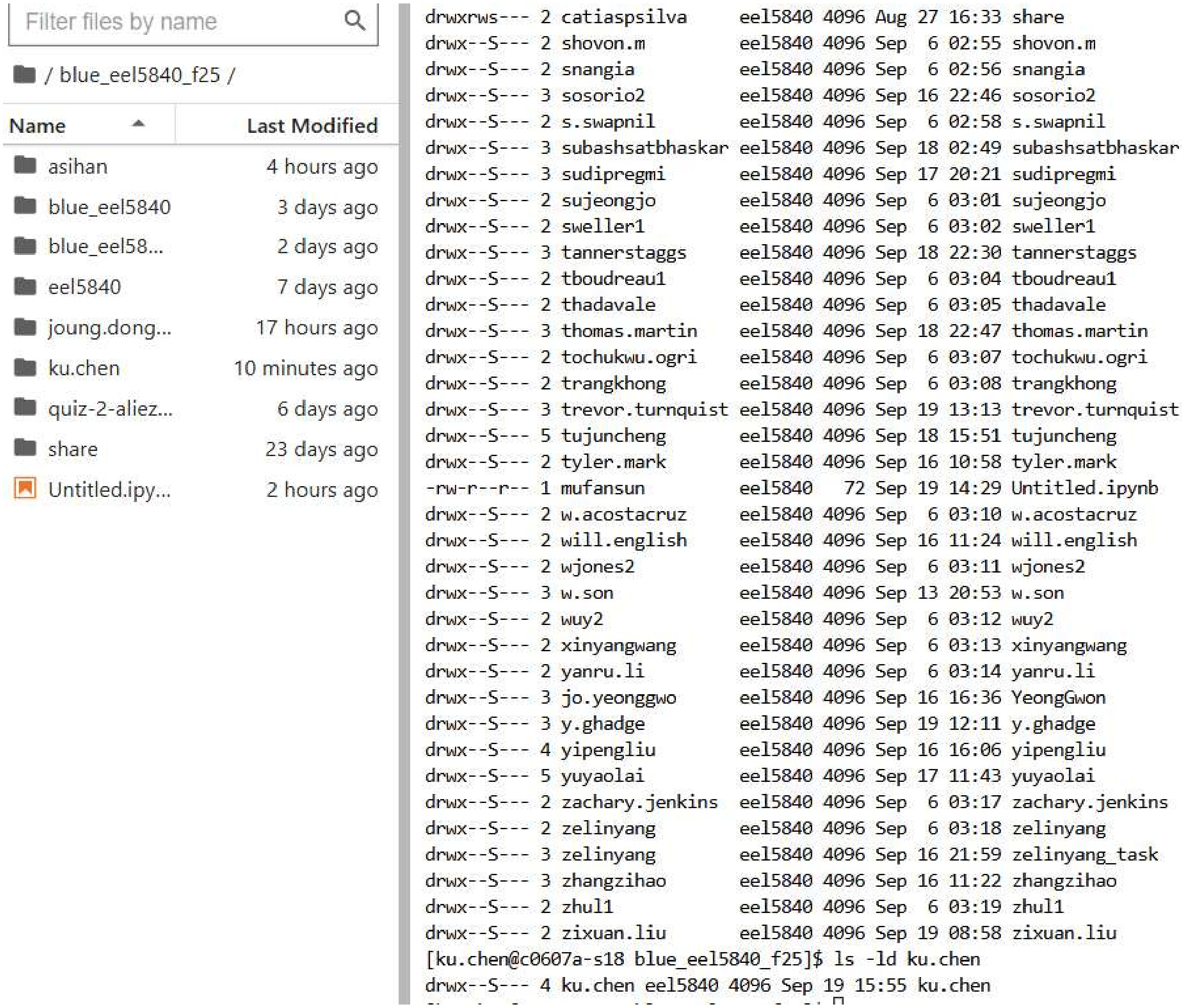
2. $0.5points$ **Create a symbolic link to map the class blue directory in your homepage. Attach a screenshot to show that this link has been created. Example and instructions below.**

```
Image('figures/symbolic_link.png', width=900)
```



3. $1 point$ **Using a terminal shell, change the directory to the _eel5840_ group. Using my naming convention, you would run ```cd blue_eel5840_f25/```. Next, if you don't see a folder with your GatorLink username, go ahead and create one run\\\\$mkdir folder_name$\\\\. Finally make sure your folder is only accessible to you by changing its mode to 700 run\\\\$chmod700 folder_name$\\\\. Attach a screenshot with these steps. See example and instructions below.**

```
Image('figures/chmod700.png', width=900)
```



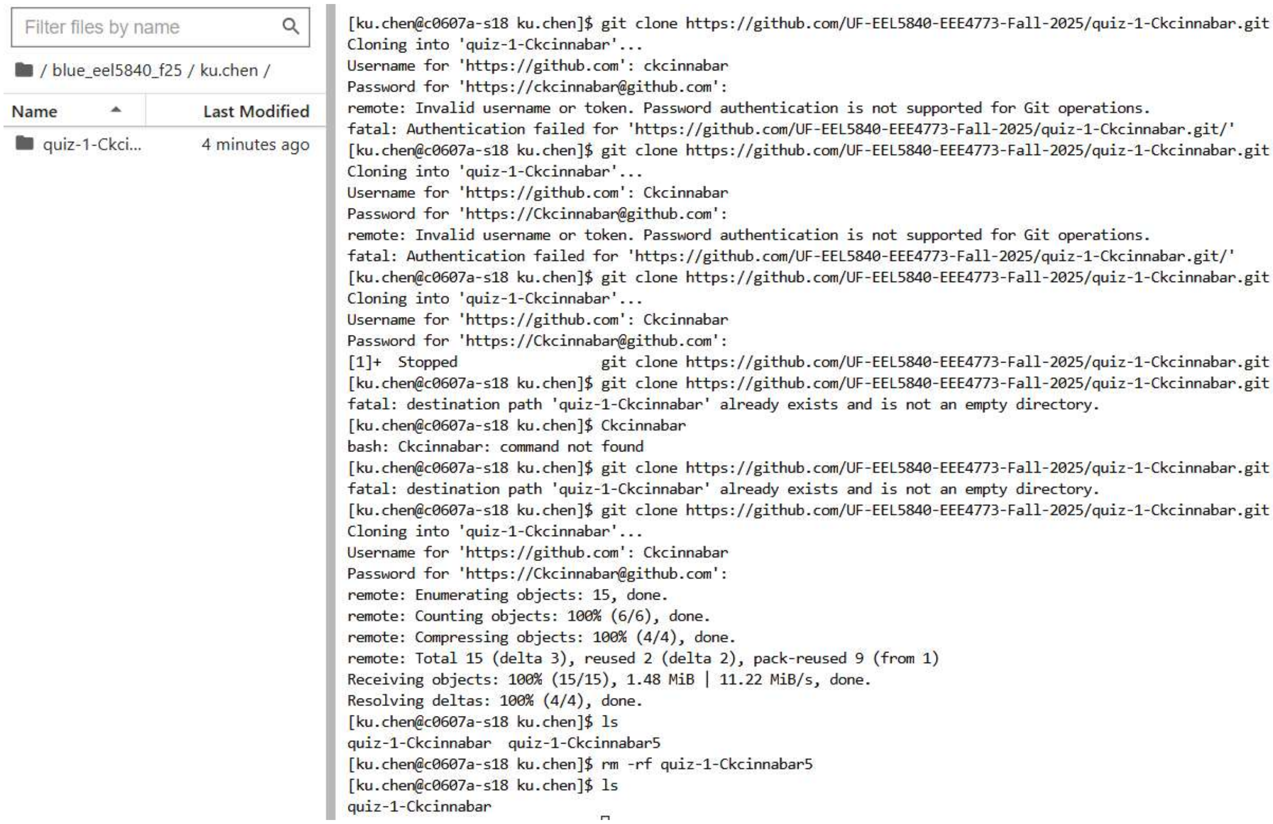
4. $1 point$ **In the terminal change the directory to your folder \\\\$cd folder_name/$\\\\, for me it's ```cd catiaspsilva/```. This is where you will clone ALL assignments. Set up your GitHub SSH key, and clone your quiz 2 repository. Attach a screenshot with this step. See example and instructions below for cloning the repository.**

For instructions on setting up your GitHub SSH key, enroll in

$$this workshop$$

$https://ufl.instructure.com/enroll/TWR9LR$**, select Module 3, and watch the 11-min video titled "Set up SSH Key" for a step-by-step hands-on demonstration.**

```
Image('figures/cloning_repo.png', width=900)
```



On-Time $1 point$

Submit your assignment before the deadline.

Submit Your Solution

Confirm that you've successfully completed the assignment.

Along with the Notebook, include a PDF of the notebook with your solutions.

```add``` and ```commit``` the final version of your work, and ```push``` your code to your GitHub repository.

Submit the URL of your GitHub Repository as your assignment submission on Canvas.
```